Numerical Methods Assignment

Total Marks: 10

Submission Deadline: Thursday, 01 October 2026

Topics: Errors, Bisection Method, Regula Falsi Method, Fixed-Point Iteration, Newton–Raphson Method, and Secant Method.

Q1. Errors **[1 Mark]**

The exact value of e is 2.71828, while its approximate value is 2.72. Calculate:

1. Absolute Error
2. Relative Error
3. Percentage Error

Q2. Bisection Method **[2 Marks]**

Using the **Bisection Method**, find an approximate root of the transcendental equation

$$e^{-x} - x = 0$$

in the interval

$$[0, 1].$$

Perform **3 iterations** and give your answer up to four decimal places.

Q3. Regula Falsi Method **[2 Marks]**

Using the **Regula Falsi Method**, find an approximate root of

$$x \log_{10}(x) - 1 = 0$$

in the interval

$$[2, 3].$$

Perform **3 iterations** and show all intermediate calculations.

Q4. Fixed-Point Iteration **[1.5 Marks]**

Consider the transcendental equation

$$x = e^{-x}.$$

take

$$x_0 = 0.5$$

and calculate x_1, x_2, and x_3.

Q5. Newton–Raphson Method **[2 Marks]**

Using the **Newton–Raphson Method**, find an approximate root of

$$x - \cos x = 0.$$

Take the initial approximation

$$x_0 = 0.5$$

and perform **3 iterations**.

Q6. Secant Method **[1.5 Marks]**

Using the **Secant Method**, find an approximate root of

$$e^x - 3x = 0$$

using the initial approximations

$$x_0 = 0, \qquad x_1 = 1.$$

Perform **3 iterations** using
Give your final answer up to four decimal places.

Total Marks = 10

Instructions: Show the formula and all intermediate calculations for each numerical method. Use a calculator in radian mode for trigonometric functions.